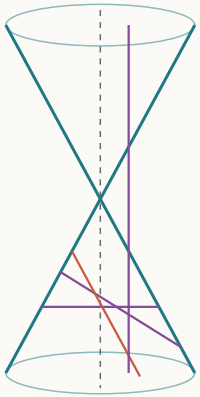
Conic sections CHEAT SHEET

SLICING A CONE



α: the cone's side to its axis
β: the cutting plane to the axis

$\beta = 90°$	circle
$\alpha < \beta < 90°$	ellipse
$\beta = \alpha$	parabola
$\beta < \alpha$	hyperbola

$e = \cos\beta / \cos\alpha$

through the tip (degenerate): a point · one line · two crossing lines

DISTANCES

$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

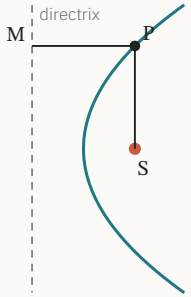
(1, 2) to (4, 6): $\sqrt{(9 + 16)} = 5$

(x, y) to the line $x = k$: $|x - k|$

THE FOUR CURVES

circle	$e = 0$	every point the same distance from the centre
ellipse	$0 < e < 1$	$PF_1 + PF_2 = 2a$
parabola	$e = 1$	PS = PM: as far from the focus as from the directrix
hyperbola	$e > 1$	$\lvert PF_1 - PF_2 \rvert = 2a$

THE PARABOLA



PS = PM:
as far from the focus S
as from the directrix

rays parallel to the axis
reflect through the focus

dish 1.2 m wide, 0.15 m deep: $0.6^2 = 4a(0.15)$, $a = 0.6$ m

FOUR STANDARD FORMS

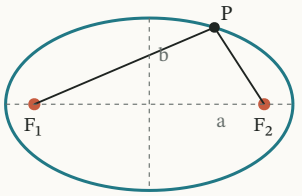
equation	focus	directrix	axis	opens
$y^2 = 4ax$	(a, 0)	$x = -a$	$y = 0$	right
$y^2 = -4ax$	(−a, 0)	$x = a$	$y = 0$	left
$x^2 = 4ay$	(0, a)	$y = -a$	$x = 0$	up
$x^2 = -4ay$	(0, −a)	$y = a$	$x = 0$	down

a > 0 · the unsquared variable names the axis · the sign says which way

PARTS OF A PARABOLA

vertex (0, 0) · axis $y = 0$ · focus (a, 0)
latus rectum: ends (a, ±2a), length 4a
focal distance of (x, y): $x + a$
parametric: $(at^2, 2at)$
$y^2 = 12x$: $a = 3$, focus (3, 0), directrix $x = -3$
$y = x^2$: $x^2 = 4(¼)y$, focus (0, ¼)
$y^2 = 8x$, the point (8, 8): focal distance $8 + 2 = 10$
vertex (0, 0), focus (0, −3): $x^2 = -12y$
symmetric about the y axis, through (4, 2): $x^2 = 8y$

THE ELLIPSE



a pencil in a loop of string
round two pins: the distances
always add to 2a

$b^2 + c^2 = a^2$ (the triangle)

larger denominator = a^2: it names the major axis

ELLIPSE FORMULAS

$PF_1 + PF_2 = 2a$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$b^2 = a^2 - c^2$ foci $(\pm c, 0)$ vertices $(\pm a, 0)$

$e = \frac{c}{a}$ $(0 \le e < 1)$ latus rectum $\frac{2b^2}{a}$

focal distances $a \pm ex$ directrices $x = \pm\frac{a}{e}$

parametric $(a\cos\theta, b\sin\theta)$ area πab

WORKED EXAMPLES

$\frac{x^2}{25} + \frac{y^2}{9} = 1$: $c = 4$, $e = \frac{4}{5}$, latus rectum $\frac{18}{5}$

$\frac{x^2}{4} + \frac{y^2}{9} = 1$ (upright): $c = \sqrt{5}$, foci $(0, \pm\sqrt{5})$

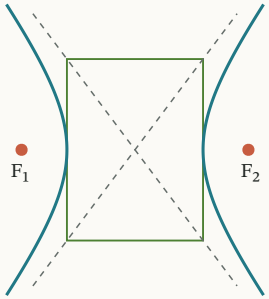
foci $(\pm 3, 0)$, vertices $(\pm 5, 0)$: $\frac{x^2}{25} + \frac{y^2}{16} = 1$

foci $(\pm 2, 0)$, $e = ½$: $a = 4$, $\frac{x^2}{16} + \frac{y^2}{12} = 1$

through (4, 3) and (6, 2): $\frac{x^2}{52} + \frac{y^2}{13} = 1$

Conic sections CHEAT SHEET

THE HYPERBOLA



the box 2a × 2b:
its diagonals are
the asymptotes

positive term names the axis

no points between x = −a and x = a

HYPERBOLA FORMULAS

$|PF_1 - PF_2| = 2a$

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$c^2 = a^2 + b^2$ foci $(\pm c, 0)$ vertices $(\pm a, 0)$

asymptotes $y = \pm\frac{b}{a}x$

$e = \frac{c}{a} > 1$ latus rectum $\frac{2b^2}{a}$

focal distances $|ex \pm a|$ directrices $x = \pm\frac{a}{e}$

parametric $(a \sec\theta, b \tan\theta)$

RECTANGULAR HYPERBOLA · EXAMPLES

$a = b$: $x^2 - y^2 = a^2$, $e = \sqrt{2}$, asymptotes $y = \pm x$

turned 45°: $xy = c^2$, parametric $(ct, c/t)$

$xy = 8$: vertex $(2\sqrt{2}, 2\sqrt{2})$, foci $(4, 4), (-4, -4)$

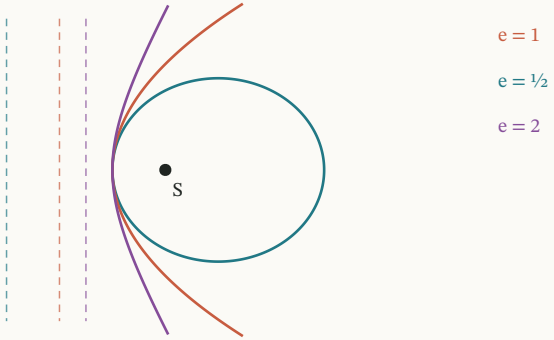
$\frac{x^2}{9} - \frac{y^2}{16} = 1$: $c = 5$, $e = \frac{5}{3}$, $y = \pm\frac{4}{3}x$

$\frac{y^2}{9} - \frac{x^2}{16} = 1$ (upright): foci $(0, \pm 5)$, $y = \pm\frac{3}{4}x$

foci $(\pm 5, 0)$, vertices $(\pm 4, 0)$: $\frac{x^2}{16} - \frac{y^2}{9} = 1$

towers 300 km apart, 60 km difference: $\frac{x^2}{900} - \frac{y^2}{21\,600} = 1$

ONE RULE FOR EVERY CONIC

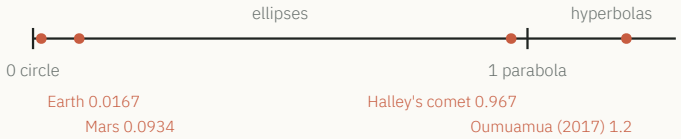


$PS = e \cdot PM$ distance to the focus = e × distance to the directrix

ECCENTRICITY

focus $(ae, 0)$, directrix $x = \frac{a}{e}$: $\frac{x^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1$

focus $(1, 0)$, directrix $x = 4$, $e = ½$: $\frac{x^2}{4} + \frac{y^2}{3} = 1$



POLAR FORM

$r = \frac{ed}{1 + e\cos\theta}$ (focus at the origin, directrix $x = d$)

$r = \frac{6}{2 + \cos\theta}$: $e = ½$, r from 2 to 6, $a = 4$, $c = 2$

nearest $a(1 - e)$, furthest $a(1 + e)$

PARAMETRIC FORMS

parabola $y^2 = 4ax$	$(at^2, 2at)$
ellipse	$(a\cos\theta, b\sin\theta)$
hyperbola	$(a\sec\theta, b\tan\theta)$
$xy = c^2$	$(ct, c/t)$

$\cos^2\theta + \sin^2\theta = 1$ · $\sec^2\theta - \tan^2\theta = 1$

TANGENTS

	slope m	at a point
parabola	$y = mx + \frac{a}{m}$	$ty = x + at^2$
ellipse	$y = mx \pm \sqrt{(a^2m^2 + b^2)}$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
hyperbola	$y = mx \pm \sqrt{(a^2m^2 - b^2)}$	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

a tangent meets the curve in a repeated root
hyperbola: only slopes steeper than ± b/a

$y^2 = 8x$, slope 2: $y = 2x + 1$, touches (½, 2)

$\frac{x^2}{25} + \frac{y^2}{9} = 1$ at $(4, 1.8)$: $4x + 5y = 25$

$\frac{x^2}{9} - \frac{y^2}{16} = 1$ at $(5, \frac{16}{3})$: $5x - 3y = 9$

THE GENERAL EQUATION

$x^2 - 6x = (x - 3)^2 - 9$

$9x^2 + 4y^2 - 36x + 24y + 36 = 0 \rightarrow \frac{(x-2)^2}{4} + \frac{(y+3)^2}{9} = 1$

$y^2 - 4y - 8x + 28 = 0 \rightarrow (y - 2)^2 = 8(x - 3)$

$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$

$B^2 - 4AC < 0$	**ellipse**	$x^2 + xy + y^2 = 3$
$B^2 - 4AC = 0$	**parabola**	$x^2 + 2xy + y^2 = 4x$
$B^2 - 4AC > 0$	**hyperbola**	$xy = 8$

watch for collapsed cases: $x^2 - y^2 = 0$ is two lines